

A Roundabout

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1 Introduction

Here we consider a generally covariant formulation of a classical mechanical problem you may be familiar with from the first year. Suppose you have taken up work as a babysitter for your strange uncle David's two children, Alice and Bob. They nag you to take them to the local playground, a request to which you eventually cave. When you arrive, Alice and Bob take up antipodal positions (that is, opposite positions) on the perimeter of a circular roundabout. Alice has a ball that she throws directly (in the $-r$ direction) towards her brother. Knowing little about classical mechanics, she expects Bob to catch it. Here, we will model the ball's path independent of coordinate system.

2 The Model

In the standard free space case, we expect a free particle along a curve $x^\mu = (x, y, t)$ to be treated as:

$$\frac{d^2 x^\mu}{d\tau^2} = 0 \quad (1)$$

This is valid in the case of a coordinate system with static basis vectors. However, in our situation, the problem from the perspective of the children takes into account *coordinate* acceleration. To consider the logic behind this, image a standard rectilinear coordinate system endowed with a field of vectors of constant length and direction. When asking of the change in the vector field in any direction, we would expect the answer to always be zero. I.e.:

$$\partial_\mu V^\nu = 0 \quad (2)$$

However, if we were to instead lay down a non-trivial coordinate system, say a curvilinear one, we will find some kind of non-zero value for the above expression due to the change in basis vectors from point to point. To counteract this, we bring in what's called the *covariant* derivative, a derivative that transforms with the coordinate system to maintain that the vector field (being a tensor field) does not change. To do this, we bring in an extra term called the *connection* which cancels the change:

$$D_\mu V^\nu = \partial_\mu V^\nu + \Gamma_{\gamma\mu}^\nu V^\gamma \quad (3)$$

This covariant change in the field remains zero:

$$D_\mu V^\nu = 0 \quad (4)$$

You can see this as a kind of "tensor" derivative.

So, in this situation, it may be tempting to treat the ball using (1), but we instead need to set the covariant acceleration to zero. Consider our path $x^\mu(t) = (r(t), \theta(t), t)$, and then consider the ball's observable tangent vector along that path:

$$v^\mu = \frac{dx^\mu}{dt} \quad (5)$$

We then consider the covariant change in this vector, recall that this is because we need to account for coordinate acceleration:

$$D_t v^\mu = \frac{dv^\mu}{dt} + \Gamma_{\alpha\beta}^\mu v^\alpha v^\beta \quad (6)$$

Written out in terms of x^μ , we have:

$$\frac{d^2 x^\mu}{dt^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{dt} \frac{dx^\beta}{dt} = 0 \quad (7)$$

This is exactly what is known as the *geodesic* equation. The equation is true for the ball in all reference frame. What will change between frames is the value of the connection coefficients. You can reason that, from your perspective, since you are stationary with respect to the roundabout, your connections will be zero, so your differential equation for the ball will reduce to (1).

The line element for our cylindrical spacetime here is logically:

$$ds^2 = dr^2 + r^2 d\theta^2 - dt^2 \quad (8)$$

However, to encode rotation from the roundabout frame, we have:

$$d\theta \rightarrow d\theta + \omega dt \quad (9)$$

Where ω is the angular velocity of the roundabout. Our line element becomes:

$$ds^2 = dr^2 + r^2(d\theta + \omega dt)^2 - dt^2 \quad (10)$$

This corresponds to a naturally non static metric tensor:

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & \omega r^2 \\ 0 & \omega r^2 & \omega^2 r^2 - 1 \end{pmatrix} \quad (11)$$

The connection coefficients function normally like this, for example:

$$\frac{\partial \vec{e}_r}{\partial \theta} = \Gamma_{r\theta}^r \vec{e}_r + \Gamma_{r\theta}^\theta \vec{e}_\theta \quad (12)$$

It essentially gives us a description as to how basis vectors shift. To calculate these in terms of our metric, we have:

$$\Gamma_{\nu\rho}^{\mu} = \frac{1}{2}g^{\mu\sigma}(\partial_{\nu}g_{\sigma\rho} + \partial_{\rho}g_{\sigma\nu} - \partial_{\sigma}g_{\nu\rho}) \quad (13)$$

This may look rather confusion, but can be easily derived by minimising the action of the straight line Lagrangian:

$$\mathcal{L} = \frac{1}{2}g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu} \quad (14)$$

The connections in our case (which are symmetric in the lower indices) are:

$$\Gamma_{\theta\theta}^r = -r \quad (15)$$

$$\Gamma_{tt}^r = -r\omega^2 \quad (16)$$

$$\Gamma_{\theta t}^r = -r\omega \quad (17)$$

$$\Gamma_{r\theta}^{\theta} = 1/r \quad (18)$$

$$\Gamma_{rt}^{\theta} = \omega/r \quad (19)$$

Let us look first at our r component of (7):

$$\ddot{r} + \Gamma_{\theta\theta}^r\dot{\theta}^2 + \Gamma_{tt}^r + \Gamma_{\theta t}^r\dot{\theta} + \Gamma_{t\theta}^r\dot{\theta} = 0 \quad (20)$$

Which expands to:

$$\ddot{r} - r\dot{\theta}^2 - r\omega^2 - 2r\omega\dot{\theta} = \ddot{r} - r(\dot{\theta} + \omega)^2 = 0 \quad (21)$$

Our θ component then is:

$$\ddot{\theta} + \Gamma_{r\theta}^{\theta}\dot{r}\dot{\theta} + \Gamma_{\theta r}^{\theta}\dot{r}\dot{\theta} + \Gamma_{rt}^{\theta}\dot{r} + \Gamma_{tr}^{\theta}\dot{r} = 0 \quad (22)$$

Which expands to:

$$\ddot{\theta} + \frac{2\dot{r}\dot{\theta}}{r} + \frac{2\omega\dot{r}}{r} = \ddot{\theta} + \frac{2\dot{r}}{r}(\dot{\theta} + \omega) = 0 \quad (23)$$

We collect our full equations of motion in the roundabout frame as:

$$\ddot{\theta} + \frac{2\dot{r}}{r}(\dot{\theta} + \omega) = 0 \quad (24)$$

$$\ddot{r} - r(\dot{\theta} + \omega)^2 = 0 \quad (25)$$

The presence of the ω term corresponds to what we typically refer to as "fictitious" forces, forces that can be entirely transformed away by coordinate changes, such as the Coriolis force and the centrifugal force. The trajectory we observe in the roundabout frame is roughly represented in Figure 1.

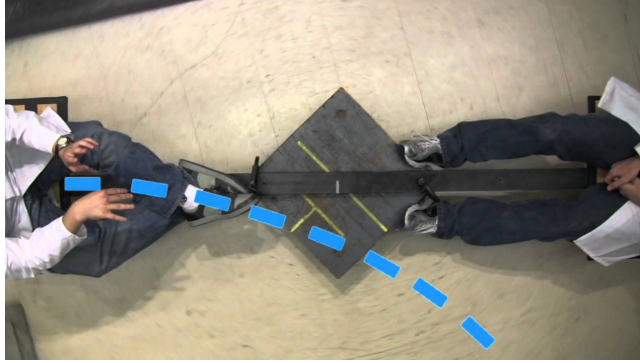


Figure 1: The roundabout frame, MIT https://www.youtube.com/watch?v=dt_XJp77-mk

Suppose now we return to your frame of reference, a nice Cartesian coordinate system represented by the flat Minkowski metric in our $2 + 1$ dimensional world. Our connections are then trivial. The equations of motion for the ball are then simply:

$$\ddot{x} = 0 \quad (26)$$

$$\ddot{y} = 0 \quad (27)$$

This trajectory is roughly represented in Figure 2, a straight line, as expected.

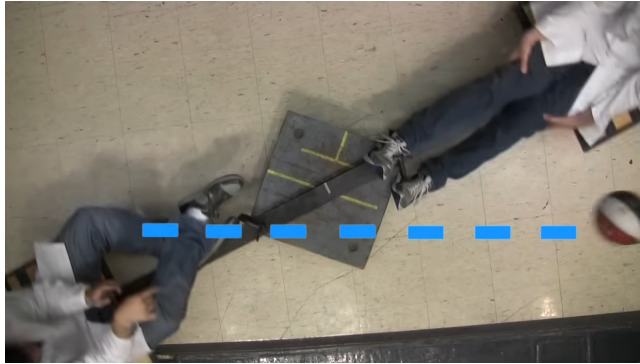


Figure 2: The stationary frame, MIT https://www.youtube.com/watch?v=dt_XJp77-mk

You have now found a covariant expression for the path of an object that is invariant of perspective. This is one of the core tenets of general relativity and modern physics. As we can see, the mathematics behind this do not have to be used in advanced or otherwise abstract circumstances, and can otherwise be explored in simple, everyday contexts.